

NAME

kh – compute the Henry constant in water or heavy water.

LIBRARY

iapws library (*libiapws*, *-liapws*)

SYNOPSIS

```

Fortran      pure subroutine Kw(T,rhow,k)
               real(dp), intent(in) :: T(:)
               real(dp), intent(in) :: rhow(:)
               real(dp), intent(out) :: k(:)

C            void iapws_r1124_Kw(size_t N,double *T,double *rhow,double *k)

Python       Kw(T:np.ndarray,rhow:np.ndarray)->Union[np.ndarray,float]:

```

DESCRIPTION

Compute the ionization constant of water K_w .

Arguments:

$T(:)$	Temperature in K.
$rhow$	Mass density in g.cm^{-3} .
$k(:)$	Ionisation constant.

Validity range:

- $273.13 \text{ K} \leq T \leq 1273.15 \text{ K}$
- $0 \leq p \leq 1000 \text{ MPa}$.

RETURN VALUE

k is filled with NaN if out of validity range.

EXAMPLES

Fortran:

```

use iapws
real(dp) :: T(1), k(1), rhow(1)
T(1) = 25.0_dp + 273.15_dp
rhow(1) = 1.0_dp
call Kw(T, rhow, 0, k)

```

C:

```

#include "iapws.h"
double T = 25.0 + 273.15;
double rhow = 1.0;
double k;
void iapws_r1124_Kw(1, &T, &rhow, &k)

```

Python:

```
import numpy as np
import pyiapws
T = np.asarray([25.0])
rhow = np.asarray([1.0])
k = pyiapws.Kw(T+273.15, rhow)
print(k)
```

SEE ALSO

**iapws(1) iapws_lib(3) iapws_psat(3) iapws_Tsat(3) iapws_kh(3) iapws_kd(3) iapws_kw(3) ciaaw(3),
codata(3),**